Tool Execution Log

A sequential record of all the tool calls and data manipulation that was done for the Honours project of Stephan Bester (27007030) at Stellenbosch University in 2026.

Aim 1: Identify species present in sample

Initial quality check

```
fastqc *.gz -o fastqc_output

cd fastqc_output

multiqc .
```

Assemble the mixed reads to obtain contigs

```
spades.py \
        --careful \
        -1 XKULL_20260212_A00904_IL30481-001_N3UD-A06_L002_R1_trimmed.fastq.gz \
        -2 XKULL_20260212_A00904_IL30481-001_N3UD-A06_L002_R2_trimmed.fastq.gz \
        -o initial_spades_output/ \
        quast.py -o QC/quast/contigs.fasta
```

Estimate completeness and contamination

```
checkm2 predict --debug --force --threads 16 --input contigs.fasta --output-directory QC/checkM2
```

Compare assembly to GTDB to obtain species predictions

```
skani search \
    -o skani_output/skani_result.txt \
    --detailed \
    --ci \
    -d /Applications/biotools/gtdb_skani_database_ani  \
    -q initial_spades_output/contigs.fasta
```

Aim 2: Obtain strain specific draft genomes

Reference genomes for *Megasphaera hutchinsoni* and *Stutzerimonas stutzeri* (GCF_001553405.1 & GCF_019704535.1) were downloaded based off the species identified in Aim 1.

Combine ref genomes for competitive mapping

The names of scaffolds were manually changed to be easier to manually identify by removing accession number. Competitive mapping was used to minimize cross-mapping between species for the initial species draft genomes.

```
cat m_hutchinsoni_reference.fna s_stutzeri_reference.fna > combined_reference.fna
```

Map mixed reads to composite reference to separate into different species

```
bowtie2-build combined_reference.fna combined_index

bowtie2 \

        --very-sensitive \

        -x combined_index \

        -1 ../stephan_read_data/XKULL_20260212_A00904_IL30481-001_N3UD-A06_L002_R1_trimmed.fastq.gz \

        -2 ../stephan_read_data/XKULL_20260212_A00904_IL30481-001_N3UD-A06_L002_R2_trimmed.fastq.gz \

        -S mapped.sam
```

Process SAM file

```
samtools view -b mapped.sam > initial_comp_mapped.bam

samtools sort initial_comp_mapped.bam -o initial_comp_mapped_sorted.bam

samtools index initial_comp_mapped_sorted.bam

samtools idxstats initial_comp_mapped_sorted.bam
```

Filter mapped reads

Only including those that mapped with MAPQ > 10.

```
samtools view \

        -b \

        -q 10 initial_comp_mapped_sorted.bam \
```

```
    -o initial_comp_mapped_sorted_q10.bam
samtools index initial_comp_mapped_sorted_q10.bam
```

Extract species reads:

```
samtools view -b initial_comp_mapped_sorted_q10.bam Stutzerimonas_stutzeri > s_stut_q10.bam
samtools index s_stut_q10.bam
```

Manually created bed file from samtools idxstats output, called “m_hutch_scaffolds.bed “.

```
samtools view \
    -b initial_comp_mapped_sorted_q10.bam \
    -L m_hutch_scaffolds.bed > m_hutch_q10.bam
samtools index m_hutch_q10.bam
```

Convert uniquely mapped reads to sequence data (fasta)

Singletons are captured explicitly to ensure equal number of forward and reverse reads are maintained.

```
samtools sort -n m_hutch_q10.bam -o m_hutch_q10_namesorted.bam
samtools fastq \
        -1 m_hutch_q10_first_R1.fastq.gz \
        -2 m_hutch_q10_first_R2.fastq.gz \
        -s m_hutch_q10_first_singletons.fastq.gz \
        m_hutch_q10_namesorted.bam
samtools sort -n s_stut_q10.bam -o s_stut_q10_namesorted.bam
samtools fastq \
        -1 s_stut_q10_first_R1.fastq.gz \
        -2 s_stut_q10_first_R2.fastq.gz \
        -s s_stut_q10_first_singletons.fastq.gz \
        s_stut_q10_namesorted.bam
```

First individual draft genome generation

```
spades.py \
        --careful \
        -1 results_with_only_single/m_hutch_q10_first_R1.fastq.gz \
```

```
        -2 results_with_only_single/m_hutch_q10_first_R2.fastq.gz \
        -s results_with_only_single/m_hutch_q10_first_singletons.fastq.gz \
        -o first_spec_assemblies/m_hutch_de_novo_careful/
spades.py \
        --careful \
        -1 results_with_only_single/s_stut_q10_first_R1.fastq.gz \
        -2 results_with_only_single/s_stut_q10_first_R2.fastq.gz \
        -s results_with_only_single/s_stut_q10_first_singletons.fastq.gz \
        -o first_spec_assemblies/s_stut_de_novo_careful/
```

The "--careful" flag was used initially to minimize mismatches. After this SPAdes recommended to use the "--isolate" flag instead. Both were used and compared.

```
spades.py \
        --isolate \
        -1 results_with_only_single/m_hutch_q10_first_R1.fastq.gz \
        -2 results_with_only_single/m_hutch_q10_first_R2.fastq.gz \
        -s results_with_only_single/m_hutch_q10_first_singletons.fastq.gz \
        -o first_spec_assemblies/m_hutch_de_novo_isolate/
spades.py \
        --isolate \
        -1 results_with_only_single/s_stut_q10_first_R1.fastq.gz \
        -2 results_with_only_single/s_stut_q10_first_R2.fastq.gz \
        -s results_with_only_single/s_stut_q10_first_singletons.fastq.gz \
        -o first_spec_assemblies/s_stut_de_novo_isolate/
```

Assess first draft genomes

For M. hutchinsoni *the "--fragmented" flag had to be used as the reference consists of 33 scaffolds.*

```
quast.py \
        first_spec_assemblies/m_hutch_de_novo_careful/contigs.fasta \
        --fragmented \
        -r ../../NCBI_Refs/m_hutchinsoni_reference.fna \
```

```
        -o quast_results/
quast.py \
        first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
        --fragmented \
        -r ../../NCBI_Refs/m_hutchinsoni_reference.fna \
        -o quast_results/
quast.py \
        first_spec_assemblies/s_stut_de_novo_careful/contigs.fasta \
        -r ../../NCBI_Refs/s_stutzeri_reference.fna \
        -o quast_results/
quast.py \
        first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        -r ../../NCBI_Refs/s_stutzeri_reference.fna \
        -o quast_results/
checkm2 predict \
        --debug \
        --threads 16 \
        --input first_spec_assemblies/m_hutch_de_novo_careful/contigs.fasta \
        --output-directory checkM2_results
checkm2 predict \
        --debug \
        --threads 16 \
        --input first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
        --output-directory checkM2_results
checkm2 predict \
        --debug \
        --threads 16 \
        --input first_spec_assemblies/s_stut_de_novo_careful/contigs.fasta \
        --output-directory checkM2_results
```

```
checkm2 predict \
        --debug \
        --threads 16 \
        --input first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        --output-directory checkM2_results
```

Compare results for “--careful" and “--isolate"

```
quast.py \
        m_hutch_de_novo_careful/contigs.fasta \
        m_hutch_de_novo_isolate/contigs.fasta \
        -t 1 \
        -r ../NCBI_Refs/m_hutchinsoni_reference.fna \
        -o m_hutch_quast_compare/
quast.py \
        s_stut_de_novo_careful/contigs.fasta \
        s_stut_de_novo_isolate/contigs.fasta \
        -t 1 \
        -r ../NCBI_Refs/m_hutchinsoni_reference.fna \
        -o s_stut_quast_compare/
skani dist \
        -q m_hutch_de_novo_careful/contigs.fasta \
        m_hutch_de_novo_isolate/contigs.fasta \
        -r ../NCBI_Refs/m_hutchinsoni_reference.fna \
        --detailed \
        --ci \
        -o skani/m_hutch
```

The results for both flags were very similar with the “--isolate" results being slightly better. Thus, the results from “--isolate" was used.

Extract reads unused reads

As looking into how to filter for reads with a MAPQ score below a certain value, the "-U" flag was found that captures all the filtered out reads. This had to be done explicitly this time but will be incorporated in the initial filtering moving forward.

```
samtools view \
        -b \
        -q 10 \
        -o delete_me.bam \
        -U initial_comp_mapped_sorted_under_q10.bam \
        initial_comp_mapped_sorted.bam
```

Convert previously unused read names to sequence data (fasta)

```
samtools sort \
        -n initial_comp_mapped_sorted_under_q10.bam \
        -o initial_comp_mapped_sorted_under_q10_namesorted.bam
samtools fastq \
        -1 unused_reads_1_R1.fastq.gz \
        -2 unused_reads_1_R2.fastq.gz \
        -s unused_reads_1_singletons.fastq.gz \
        initial_comp_mapped_sorted_under_q10_namesorted.bam
```

Map unused reads to individual species assemblies

Individual mapping was used from here on to capture as many reads as possible by mapping against the draft genomes that were assembled conservatively via competitive mapping.

```
bowtie2-build \
        ../../comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
        m_hutch_first_assembly_index
bowtie2 \
        --very-sensitive \
        -x m_hutch_first_assembly_index \
        -1 ../../comp_map_initial/results_with_only_single/unused_reads_1_R1.fastq.gz \
```

```
        -2 ../../comp_map_initial/results_with_only_single/unused_reads_1_R2.fastq.gz \
        -U ../../comp_map_initial/results_with_only_single/unused_reads_1_singletons.fastq.gz \
        -S m_hutch_first_idv_map.sam
samtools view -b m_hutch_first_idv_map.sam > m_hutch_first_idv_map.bam
samtools sort m_hutch_first_idv_map.bam -o m_hutch_first_idv_map_sorted.bam
samtools index m_hutch_first_idv_map_sorted.bam
bowtie2-build \
        ../../comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        s_stut_first_assembly_index
bowtie2 --very-sensitive \
        -x s_stut_first_assembly_index \
        -1 ../../comp_map_initial/results_with_only_single/unused_reads_1_R1.fastq.gz \
        -2 ../../comp_map_initial/results_with_only_single/unused_reads_1_R2.fastq.gz \
        -U ../../comp_map_initial/results_with_only_single/unused_reads_1_singletons.fastq.gz \
        -S s_stut_first_idv_map.sam
samtools view -b s_stut_first_idv_map.sam > s_stut_first_idv_map.bam
samtools sort s_stut_first_idv_map.bam -o s_stut_first_idv_map_sorted.bam
samtools index s_stut_first_idv_map_sorted.bam
samtools idxstats s_stut_first_idv_map_sorted.bam > idxstats_s_stut_first_idv_map_sorted.txt
```

Filter mapped reads

Only including those that mapped with MAPQ > 10, capturing the filtered out reads.

```
samtools view \
        -b \
        -q 10 \
        -o m_hutch_first_idv_map_sorted_q10.bam \
        -U m_hutch_first_idv_map_sorted_under_q10.bam \
        m_hutch_first_idv_map_sorted.bam
samtools view \
```

```
    -b \
    -q 10 \
    -o s_stut_first_idv_map_sorted_q10.bam \
    -U s_stut_first_idv_map_sorted_under_q10.bam \
    s_stut_first_idv_map_sorted.bam
```

Combine previously used and newly mapped reads

```
samtools merge \
    m_hutch_reads_round_2.bam \
    ../m_hutch_first_idv_map_sorted_q10.bam \
    ../../../comp_map_initial/results_with_only_single/m_hutch_q10_namesorted.bam
samtools merge \
    s_stut_reads_round_2.bam \
    ../s_stut_first_idv_map_sorted_q10.bam \
    ../../../comp_map_initial/results_with_only_single/s_stut_q10_namesorted.bam
```

Convert updated species read sets to sequence data (fasta)

```
samtools sort -n m_hutch_reads_round_2.bam -o m_hutch_reads_round_2_namesorted.bam
samtools fastq \
    -1 m_hutch_q10_second_R1.fastq.gz \
    -2 m_hutch_q10_second_R2.fastq.gz \
    -s m_hutch_q10_second_singletons.fastq.gz \
    m_hutch_reads_round_2.bam
samtools sort -n s_stut_reads_round_2.bam -o s_stut_reads_round_2_namesorted.bam
samtools fastq \
    -1 s_stut_q10_second_R1.fastq.gz \
    -2 s_stut_q10_second_R2.fastq.gz \
    -s s_stut_q10_singletons.fastq.gz \
    s_stut_reads_round_2_namesorted.bam
```

Reassemble draft genomes

```
spades.py \
        --isolate \
        -1 m_hutch_q10_second_R1.fastq.gz \
        -2 m_hutch_q10_second_R2.fastq.gz \
        -s m_hutch_q10_second_singletons.fastq.gz \
        -o ./
spades.py \
        --isolate \
        -1 s_stut_q10_second_R1.fastq.gz \
        -2 s_stut_q10_second_R2.fastq.gz \
        -s s_stut_q10_singletons.fastq.gz \
        -o ./
```

Evaluate new draft genomes

```
quast.py \
        comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
        mapping_round_2/m_hutch/assembly/contigs.fasta \
        -t 1 \
        --fragmented \
        -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
        -o mapping_round_2/m_hutch/qc/quast/
checkm2 predict \
        --debug \
        --threads 16 \
        --input assembly/contigs.fasta \
        --output-directory qc/checkM2/
skani dist \
        -q comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
```

```
        mapping_round_2/m_hutch/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
        --detailed \
        --ci \
        -o mapping_round_2/m_hutch/qc/skani_results
quast.py \
        comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        mapping_round_2/s_stut/assembly/contigs.fasta \
        -t 1 \
        --fragmented \
        -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
        -o mapping_round_2/s_stut/qc/quast
checkm2 predict \
        --debug \
        --threads 16 \
        --input assembly/contigs.fasta \
        --output-directory qc/checkM2
skani dist \
        -q comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        mapping_round_2/s_stut/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
        --detailed \
        --ci \
        -o mapping_round_2/s_stut/qc/s_stut_skani
```

Assemble third iteration of draft genomes

At this stage it was realised that using a pipeline for this would be much more efficient, but the naming conventions used made creating a pipeline troublesome. Thus, everything was repeated to standardise naming conventions.

Extract unique filtered out read names for each species set

```
samtools view \
        m_hutch_first_idv_map_sorted_under_q10.bam \
        | cut -f1 \
        | sort -u \
        > m_hutch_unmapped_reads.txt
samtools view \
        s_stut_first_idv_map_sorted_under_q10.bam \
        | cut -f1 \
        | sort -u \
        > s_stut_unmapped_reads.txt
```

Extract reads that were filtered out from both sets

```
comm -12 m_hutch_unmapped_reads.txt s_stut_unmapped_reads.txt > unmapped_reads.txt
```

Extract sequence alignment data for reads filtered out of both sets

```
samtools view \
        -N unmapped_reads.txt \
        -b ../../comp_map_initial/results_with_only_single/initial_comp_mapped_sorted.bam \
        > unmapped_reads_round_3.bam
```

Convert to sequence data (fasta)

```
samtools sort -n unmapped_reads_round_3.bam -o unmapped_reads_round_3_namesorted.bam
samtools fastq \
        -1 unmapped_round_3_R1.fastq.gz \
        -2 unmapped_round_3_R2.fastq.gz \
        -s unmapped_round_3_singletons.fastq.gz \
        unmapped_reads_round_3_namesorted.bam
```

Map previously unused reads to current draft genomes

```
bowtie2-build \
        ../../mapping_round_2/m_hutch/assembly/contigs.fasta \
        m_hutch_from_round_2_index
```

```
bowtie2 \
        --very-sensitive -x m_hutch_from_round_2_index \
        -1 ../reads_from_round_2/unmapped_round_3_R1.fastq.gz \
        -2 ../reads_from_round_2/unmapped_round_3_R2.fastq.gz \
        -U ../reads_from_round_2/unmapped_round_3_singletons.fastq.gz \
        -S m_hutch_round_3.sam
bowtie2-build \
        ../../mapping_round_2/s_stut/assembly/contigs.fasta \
        s_stut_from_round_2_index
bowtie2 \
        --very-sensitive -x s_stut_from_round_2_index \
        -1 ../reads_from_round_2/unmapped_round_3_R1.fastq.gz \
        -2 ../reads_from_round_2/unmapped_round_3_R2.fastq.gz \
        -U ../reads_from_round_2/unmapped_round_3_singletons.fastq.gz \
        -S s_stut_round_3.sam
```

Process SAM files

```
samtools view -b m_hutch_round_3.sam > m_hutch_round_3.bam
samtools sort m_hutch_round_3.bam -o m_hutch_round_3_sorted.bam
samtools index m_hutch_round_3_sorted.bam
samtools view -b s_stut_round_3.sam > s_stut_round_3.bam
samtools sort s_stut_round_3.bam -o s_stut_round_3_sorted.bam
samtools index s_stut_round_3_sorted.bam
```

Filter newly mapped reads

```
samtools view \
        -b \
        -q 10 \
        -o m_hutch_round_3_sorted_q10.bam \
        -U m_hutch_round_3_sorted_under_q10.bam \
        m_hutch_round_3_sorted.bam
```

```
samtools view \
        -b \
        -q 10 \
        -o s_stut_round_3_sorted_q10.bam \
        -U s_stut_round_3_sorted_under_q10.bam \
        s_stut_round_3_sorted.bam
```

Combine previously used and new reads for each species

```
samtools merge \
        m_hutch_reads_for_third_assembly.bam \
        m_hutch_round_3_sorted_q10.bam \
        ../../mapping_round_2/m_hutch/assembly/m_hutch_reads_round_2.bam
samtools merge \
    s_stut_for_third_assembly.bam \
    s_stut_round_3_sorted_q10.bam \
    ../../mapping_round_2/s_stut/assembly/s_stut_reads_round_2.bam
```

Convert to sequence data (fasta)

```
samtools sort \
        -n m_hutch_reads_for_third_assembly.bam \
        -o m_hutch_reads_for_third_assembly_namesorted.bam
samtools fastq \
    -1 m_hutch_3_R1.fastq.gz \
    -2 m_hutch_3_R2.fastq.gz \
    -s m_hutch_3_singletons.fastq.gz m_hutch_reads_for_third_assembly_namesorted.bam
samtools sort \
    -n s_stut_for_third_assembly.bam \
    -o s_stut_for_third_assembly_namesorted.bam
samtools fastq \
    -1 s_stut_3_R1.fastq.gz \
    -2 s_stut_3_R2.fastq.gz \
```

```
    -s s_stut_3_singletons.fastq.gz s_stut_for_third_assembly_namesorted.bam
```

Assemble third draft genomes

```
spades.py --isolate -1 ../m_hutch_3_R1.fastq.gz -2 ../m_hutch_3_R2.fastq.gz -s
../m_hutch_3_singletons.fastq.gz -o ./
```

```
spades.py --isolate -1 s_stut_3_R1.fastq.gz -2 s_stut_3_R2.fastq.gz -s s_stut_3_singletons.fastq.gz -
o assembly/
```

Evaluate and compare new draft genomes

```
quast.py \
    ../../comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
    ../../mapping_round_2/m_hutch/assembly/contigs.fasta \
    Assembly/contigs.fasta \
    -t 1 \
    --fragmented \
    -r ../../comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
    -o qc/
quast.py \
    ../../comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
    ../../mapping_round_2/s_stut/assembly/contigs.fasta \
    assembly/contigs.fasta \
    -t 1 \
    -r ../../comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
    -o qc/quast/
checkm2 predict \
    --debug \
    --threads 16 \
    --input assembly/contigs.fasta \
    --output-directory qc/checkM2/
checkm2 predict \
    --debug \
```

```
    --threads 16 \
    --input Assembly/contigs.fasta \
    --output-directory qc/checkM2/
```

Iterative draft genome assembly pipeline

A pipeline called "improve_assemblies_pipeline,sh" (using the same methods previously described) was used to generate 12 more iterations of draft genomes.

```
./improve_assemblies_pipeline.sh |& tee improve_assemblies_pipeline_5_15.log
```

Evaluate and compare 15 draft genomes for each species

A pipeline was used to run CheckM2.

```
quast.py \
    comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
    mapping_round_2/m_hutch/assembly/contigs.fasta \
    mapping_round_3/m_hutch/Assembly/contigs.fasta \
    pipeline_mapping_round_4/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_5/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_6/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_7/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_8/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_9/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_10/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_11/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_12/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_13/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_14/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_15/m_hutch/assembly/contigs.fasta \
    -t 1 \
    --fragmented \
    -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
```

```
        -o qc_on_1_15/
quast.py \
        comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        mapping_round_2/s_stut/assembly/contigs.fasta \
        mapping_round_3/s_stut/Assembly/contigs.fasta \
        pipeline_mapping_round_4/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_5/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_6/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_7/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_8/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_9/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_10/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_11/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_12/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_13/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_14/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_15/s_stut/assembly/contigs.fasta \
        -t 1 \
        -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
        -o qc_on_1_15/s_stut/quast/
skani dist \
        -q comp_map_initial/first_spec_assemblies/m_hutch_de_novo_isolate/contigs.fasta \
        mapping_round_2/m_hutch/assembly/contigs.fasta \
        mapping_round_3/m_hutch/Assembly/contigs.fasta \
        pipeline_mapping_round_4/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_5/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_6/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_7/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_8/m_hutch/assembly/contigs.fasta \
```

```
        pipeline_mapping_round_9/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_10/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_11/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_12/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_13/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_14/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_15/m_hutch/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
        --detailed \
        --ci \
        -o qc_on_1_15/m_hutch/m_hutch_1_15_skani_results
skani dist \
        -q comp_map_initial/first_spec_assemblies/s_stut_de_novo_isolate/contigs.fasta \
        mapping_round_2/s_stut/assembly/contigs.fasta \
        mapping_round_3/s_stut/Assembly/contigs.fasta \
        pipeline_mapping_round_4/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_5/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_6/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_7/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_8/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_9/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_10/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_11/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_12/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_13/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_14/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_15/s_stut/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
        --detailed \
```

```
    --ci \
    -o qc_on_1_15/s_stut/s_stut_1_15_skani_results
./checkm2_on_pipeline.sh |& tee checkm2_on_pipeline_5_15.log
```

Draft genome iteration 16 to 20

When graphing resulting draft genome composite genome score, the results were unsatisfactory to observe the trend. Hence the pipelines were used to generate and evaluate 5 more iterations of draft assemblies.

```
./improve_assemblies_pipeline.sh |& tee improve_assemblies_pipeline_16_20.log
quast.py \
    pipeline_mapping_round_16/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_17/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_18/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_19/m_hutch/assembly/contigs.fasta \
    pipeline_mapping_round_20/m_hutch/assembly/contigs.fasta \
    -t 1 \
    --fragmented \
    -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
    -o qc_on_16_20/m_hutch/quast/
quast.py \
    pipeline_mapping_round_16/s_stut/assembly/contigs.fasta \
    pipeline_mapping_round_17/s_stut/assembly/contigs.fasta \
    pipeline_mapping_round_18/s_stut/assembly/contigs.fasta \
    pipeline_mapping_round_19/s_stut/assembly/contigs.fasta \
    pipeline_mapping_round_20/s_stut/assembly/contigs.fasta \
    -t 1 \
    -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
    -o qc_on_16_20/s_stut/quast/
skani dist \
    -q pipeline_mapping_round_16/m_hutch/assembly/contigs.fasta \
```

```
        pipeline_mapping_round_17/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_18/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_19/m_hutch/assembly/contigs.fasta \
        pipeline_mapping_round_20/m_hutch/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/m_hutchinsoni_reference.fna \
        --detailed \
        --ci \
        -o qc_on_16_20/m_hutch/m_hutch_16_20_skani_results
skani dist \
        -q pipeline_mapping_round_16/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_17/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_18/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_19/s_stut/assembly/contigs.fasta \
        pipeline_mapping_round_20/s_stut/assembly/contigs.fasta \
        -r comp_map_initial/NCBI_Refs/s_stutzeri_reference.fna \
        --detailed \
        --ci \
        -o qc_on_16_20/s_stut/s_stut_16_20_skani_results
./checkm2_on_pipeline.sh |& tee checkm2_on_pipeline_16_20.log
```

The draft genomes with the highest composite genome score for each species were selected and used for subsequent analysis

Aim 3: Identify possible cross-feeding interactions

Assign KEGG orthology terms to draft genomes

```
bakta \
        --db /Applications/biotools/bakta_db/db/ \
        --prefix m_hutch \
        --output bakta/m_hutch/ \
        --genus Megasphaera \
        --species hutchinsoni \
        --gram - \
        --keep-contig-headers \
        --threads 8 \
        Assemblies/m_hutch_best_assembly.fasta |& tee bakta/s_m_hutch_bakta_log.log
bakta \
        --db /Applications/biotools/bakta_db/db/ \
        --prefix s_stut \
        --output bakta/s_stut/ \
        --genus Stutzerimonas \
        --species stutzeri \
        --gram - \
        --keep-contig-headers \
        --threads 8 \
        Assemblies/s_stut_best_assembly.fasta |& tee bakta/s_stut_bakta_log.log
```

Generate predicted metabolic maps

The online version of BLASTKoala was used.

Annotate reference genomes

```
bakta \
        --db /Applications/biotools/bakta_db/db \
        --prefix ncbi_m_hutch \
        --output on_refs/bakta/ncbi_m_hutch \
        --genus Megasphaera \
        --species hutchinsoni \
```

```
        --gram - \
        --threads 8 \
        NCBI_refs/m_hutchinsoni_reference.fna \
        |& tee on_refs/bakta/ncbi_m_hutch/ncbi_m_hutch_log.log
bakta \
        --db /Applications/biotools/bakta_db/db \
        --prefix ncbi_s_stut \
        --output on_refs/bakta/ncbi_s_stut \
        --genus Stutzerimonas \
        --species stutzeri \
        --gram - \
        --threads 8 \
        NCBI_refs/s_stutzeri_reference.fna \
        |& tee on_refs/bakta/ncbi_s_stut/ncbi_s_stut_log.log
```

Metabolic reconstruction Predict required metabolites and transporters

```
gapseq doall -v 1 ../../bakta/m_hutch/m_hutch.faa |& tee m_hutch_gapseq.log
gapseq doall -v 1 ../../bakta/s_stut/s_stut.faa |& tee s_stut_gapseq.log
gapseq doall \
        -v 1 \
        ../../on_refs/bakta/ncbi_m_hutch/ncbi_m_hutch.faa \
        |& tee NCBI_m_hutch_gapseq.log
gapseq doall \
        -v 1 \
        ../../on_refs/bakta/ncbi_s_stut/ncbi_s_stut.faa \
        |& tee NCBI_s_stut_gapseq.log
```